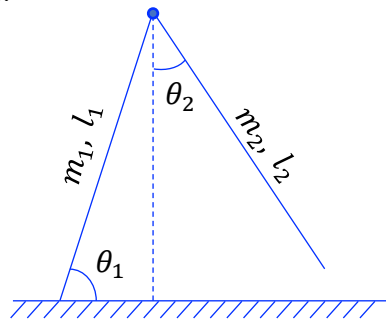


Get the equation of motion from energy method:



$$T = \frac{1}{2}J_1\dot{\theta}_1^2 + \frac{1}{2}J_2(\dot{\theta}_2)^2 + \frac{1}{2}m_2 \left( (\dot{\theta}_1 l_1 \sin \theta_1 - \dot{\theta}_2 l'_2 \cos \theta_2)^2 + (\dot{\theta}_1 l_1 \cos \theta_1 + \dot{\theta}_2 l'_2 \sin \theta_2)^2 \right)$$

$$L = T - V = \frac{1}{2}J_1\dot{\theta}_1^2 + \frac{1}{2}J_2(\dot{\theta}_2)^2 + \frac{1}{2}m_2 \left( (\dot{\theta}_1 l_1)^2 + (\dot{\theta}_2 l'_2)^2 + 2\dot{\theta}_1 l_1 \dot{\theta}_2 l'_2 \sin(\theta_2 - \theta_1) \right) - m_1 g l'_1 \sin \theta_1 - m_2 g (l_1 \sin \theta_1 - l'_2 \cos \theta_2)$$

$$\frac{\partial L}{\partial \theta_1} = -m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) - m_1 g l'_1 \cos \theta_1 - m_2 g l_1 \cos \theta_1$$

$$\frac{\partial L}{\partial \theta_2} = m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) - m_2 g l'_2 \sin \theta_2$$

$$\frac{\partial L}{\partial \dot{\theta}_1} = J_1 \dot{\theta}_1 + m_2 \dot{\theta}_1 l_1^2 + m_2 l_1 \dot{\theta}_2 l'_2 \sin(\theta_2 - \theta_1)$$

$$\frac{\partial}{\partial t} \frac{\partial L}{\partial \dot{\theta}_1} = J_1 \ddot{\theta}_1 + m_2 \ddot{\theta}_1 l_1^2 + m_2 l_1 l'_2 \ddot{\theta}_2 \sin(\theta_2 - \theta_1) + m_2 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1)$$

$$\frac{\partial L}{\partial \dot{\theta}_2} = J_2 \dot{\theta}_2 + m_2 \dot{\theta}_2 (l'_2)^2 + m_2 l_1 \dot{\theta}_1 l'_2 \sin(\theta_2 - \theta_1)$$

$$\frac{\partial}{\partial t} \frac{\partial L}{\partial \dot{\theta}_2} = J_2 \ddot{\theta}_2 + m_2 \ddot{\theta}_2 (l'_2)^2 + m_2 l_1 \ddot{\theta}_1 l'_2 \sin(\theta_2 - \theta_1) + m_2 l_1 \dot{\theta}_1 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1)$$

$$\frac{\partial}{\partial t} \frac{\partial L}{\partial \dot{\theta}_1} - \frac{\partial L}{\partial \theta_1} = J_1 \ddot{\theta}_1 + m_2 \ddot{\theta}_1 l_1^2 + m_2 l_1 l'_2 \ddot{\theta}_2 \sin(\theta_2 - \theta_1) + m_2 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) - (-m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) - m_1 g l'_1 \cos \theta_1 - m_2 g l_1 \cos \theta_1) = 0$$

$$\frac{\partial}{\partial t} \frac{\partial L}{\partial \dot{\theta}_2} - \frac{\partial L}{\partial \theta_2} = J_2 \ddot{\theta}_2 + m_2 \ddot{\theta}_2 (l'_2)^2 + m_2 l_1 \ddot{\theta}_1 l'_2 \sin(\theta_2 - \theta_1) + m_2 l_1 \dot{\theta}_1 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) - (m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) - m_2 g l'_2 \sin \theta_2) = 0$$

$$J_1 \ddot{\theta}_1 + m_2 \ddot{\theta}_1 l_1^2 + m_2 l_1 l'_2 \ddot{\theta}_2 \sin(\theta_2 - \theta_1) = -m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) - m_1 g l'_1 \cos \theta_1 - m_2 g l_1 \cos \theta_1 - m_2 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1)$$

$$J_2 \ddot{\theta}_2 + m_2 \ddot{\theta}_2 (l'_2)^2 + m_2 l_1 \ddot{\theta}_1 l'_2 \sin(\theta_2 - \theta_1) = m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) - m_2 g l'_2 \sin \theta_2 - m_2 l_1 \dot{\theta}_1 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1)$$

$$\begin{aligned}
& \begin{bmatrix} (J_1 + m_2 l_1^2) & (m_2 l_1 l_2' \sin(\theta_2 - \theta_1)) \\ (m_2 l_1 l_2' \sin(\theta_2 - \theta_1)) & J_2 + m_2 (l_2')^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \\
& = \begin{bmatrix} -m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l_2' \cos(\theta_2 - \theta_1) - m_1 g l_1' \cos \theta_1 - m_2 g l_1 \cos \theta_1 - m_2 l_1 \dot{\theta}_2 l_2' \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) \\ m_2 \dot{\theta}_1 l_1 \dot{\theta}_2 l_2' \cos(\theta_2 - \theta_1) - m_2 g l_2' \sin \theta_2 - m_2 l_1 \dot{\theta}_1 l_2' \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) \end{bmatrix}
\end{aligned}$$